SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FINAL VIVA-VOCE EXAMINATION-PRESENTATION

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Programme: B.Tech / M.Tech

Academic Year: 2025-2026

LEVERAGING THE GENETIC ALGORITHM FOR OPTIMIZING AGRICULTURAL SEED AND PRODUCT TRANSPORTATION

Project ID:

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ABSTRACT

Agricultural logistics routing involves significant computational challenges due to geographically dispersed delivery locations, sparse road connectivity, and open-route delivery requirements. Inefficient routing in such environments leads to increased travel time, fuel consumption, and operational costs. To address these challenges, this paper proposes a hierarchical optimization framework that formulates agricultural transportation routing as a Clustered Traveling Salesman Problem (CTSP). The framework integrates K-Means clustering for spatial decomposition with a Two-Level Genetic Algorithm (TLGA) that separately optimizes inter-cluster sequencing and intra-cluster routing under open-route constraints. The proposed approach is evaluated on a large-scale 1,000-node synthetic dataset and a 100-node real-world road network from the Ghaziabad region derived from OpenStreetMap, capturing realistic road topology and connectivity constraints. Experimental results demonstrate that the TLGA-based framework consistently achieves lower total travel distance compared to single-level and alternative metaheuristic approaches. The reduced routing cost directly translates to improved delivery efficiency and reduced fuel consumption, highlighting the suitability of the proposed method for real-world agricultural logistics planning.

PROJECT OVERVIEW

Agricultural logistics is an essential element of rural supply chain management, directly affecting delivery efficiency, operational costs, and product availability. In actual rural transportation networks, delivery sites are geographically scattered, and road connectivity is frequently sparse or inconsistent, rendering route planning significantly difficult.

Conventional routing methods inadequately represent these limits, leading to extended trip distances, heightened fuel consumption, and suboptimal logistical planning.

This study presents a hierarchical optimization framework that integrates clustering and evolutionary computation to produce scalable and realistic routing solutions for agricultural transportation.

Application domain:

1. Agricultural Supply chain optimization
2. Seed Distribution Networks
3. Rural product Transportations
4. Smart Logistics Planning

PROBLEM STATEMENT

The delivery of agricultural goods in rural and semi-urban areas has considerable optimization issues due to:

1. Geographically distributed agricultural operations
2. Restricted roadway access
3. Scant transportation networks
4. Unrestricted delivery routes

Current optimization methodologies presume:

1. Complete graphs
2. Euclidean metrics
3. Closed-loop navigation

These assumptions do not accurately reflect actual agricultural logistics systems.

Consequently, a scalable routing system is required to address real-world restricted transportation challenges.

OBJECTIVES

Primary Objective:

Develop an intelligent optimization framework for agricultural transportation using hierarchical genetic optimization.

Specific Objective:

- Model agricultural transportation as Clustered TSP
- Reduce routing complexity using K-Means clustering
- Optimize global and local routing using TLGA
- Validate performance on synthetic and real datasets
- Compare against existing metaheuristic techniques

LITERATURE REVIEW

Ref.	Authors	Title	Problem Addressed	Method Used	Results	Limitations
[1]	M. L. Fredman et al. (1995)	Data Structures for Traveling Salesmen	Computational complexity of TSP	Data structure and complexity analysis	Demonstrated NP-hardness and scalability limitations of TSP	No heuristic or practical solution framework
[2]	A. Adewole et al. (2011)	A Genetic Algorithm for Solving Travelling Salesman Problem	Classical TSP on Euclidean graphs	GA with permutation encoding	Achieved better solutions than greedy heuristics	Assumes fully connected Euclidean graphs
[5]	A. P. S. et al. (2013)	Open Loop Travelling Salesman Problem using Genetic Algorithm	Open-route TSP (non-return routing)	Genetic Algorithm	Demonstrated suitability of open-path routing for logistics	No clustering or real road-network modeling
[6]	D. E. Goldberg (1989)	Genetic Algorithms in Search, Optimization and Machine Learning	General optimization problems	Classical Genetic Algorithm (selection, crossover, mutation)	Established GA as a robust global optimization framework	Not specific to routing or logistics problems
[10]	S. Linganathan & P. Singamsetty (2024)	Genetic Algorithm to the Bi-objective Multiple Travelling Salesman Problem	Bi-objective MTSP (distance + workload balancing)	Multi-objective Genetic Algorithm	Improved trade-off between distance minimization and workload distribution	Assumes idealized network; no clustering or constraints

LITERATURE REVIEW

Ref.	Authors	Title	Problem Addressed	Method Used	Results	Limitations
[13]	C. Ding & G. Cheng (2007)	Two-Level Genetic Algorithm for Clustered Traveling Salesman Problem	Clustered TSP (CTSP)	Two-Level Genetic Algorithm (TLGA)	Reduced computational complexity and improved scalability	Assumes closed routes and Euclidean graph structure
[15]	A. Jaradat et al. (2019)	Solving Traveling Salesman Problem using Firefly Algorithm and K-Means Clustering	TSP optimization with clustering	K-Means + Firefly Algorithm	Improved routing efficiency via reduced search space	No hierarchical GA or constrained road-network modeling
[17]	Tan et al. (2016)	Bilevel Genetic Algorithm with Clustering for Large Scale TSP	Large-scale TSP	Bilevel GA with clustering	Reduced search space and improved scalability	No real-world road-network constraints
[18]	A. T. Al-Taani et al. (2023)	Solving the Multiple Travelling Salesman Problem using Gaussian Mixture Model and Artificial Neural Network	MTSP with clustering	GMM clustering + ANN	Competitive performance with learning-based clustering	Does not address open-route or real road-network constraints
[19]	Tan et al. (2016)	Genetic Algorithms Based on Clustering for Traveling Salesman Problems	Large-scale TSP	K-Means clustering + GA	Improved convergence speed and solution quality	Assumes fully connected graphs

RESEARCH GAP

Existing Work	Gap Identified
Traditional TSP and GA-based routing models provide efficient route optimization on Euclidean or fully connected graphs.	Lack of support for **real-world sparse road-network constraints** where direct travel between all nodes is not possible.
Clustering-based optimization techniques improve scalability for large routing problems.	Most approaches do not consider **path-aware routing with intermediate mandatory nodes** between locations.
Open-route TSP models support non-return delivery scenarios.	Existing studies do not integrate **open-route optimization with clustered hierarchical routing** for agricultural logistics.
Two-Level Genetic Algorithms improve search efficiency in clustered TSP problems.	Prior work lacks validation on **real datasets with route constraints**.

Final Statement:
These gaps serve as the motivation for the proposed system, which aims to overcome these limitations.

EXISTING SYSTEM

Current routing systems rely on:

- Classical TSP solvers
- Genetic Algorithms
- Ant Colony Optimization
- Particle Swarm Optimization

Limitations

These systems suffer from:

- Poor scalability for large datasets
- Unrealistic fully connected assumptions
- Inability to handle sparse road networks
- Lack of open-route support
- This leads to inefficient routing and increased operational costs.

PROPOSED SYSTEM

The proposed system introduces a **graph-aware hierarchical optimization framework** for agricultural transportation under real-world road constraints.

The system operates in five major stages:

1. Dataset Acquisition

For Implementation purposes the transportation node data is collected from:

- Synthetic datasets from Kaggle
- Real-world road networks using OpenStreetMap.

PROPOSED SYSTEM

2. Graph Construction & Route Discovery

Instead of assuming direct connectivity between all locations, the transportation network is modeled as a **degree-constrained graph**, where:

- Nodes represent delivery locations.
- Edges represent feasible road connections.
- The system first identifies all possible routes between connected locations.
- Intermediate junctions are considered during path traversal.
- This enables realistic route planning based on actual road availability.

3. Spatial Clustering

K-Means clustering groups geographically close nodes into clusters, reducing the global search space.

PROPOSED SYSTEM

4. Hierarchical Route Optimization

A **Two-Level Genetic Algorithm (TLGA)** performs optimization:

Macro-Level Optimization:

Determines the optimal sequence of cluster visitation.

Micro-Level Optimization:

Within each cluster: Multiple feasible paths between nodes are evaluated.

The best route is selected based on:

- Minimum travel distance
- Route feasibility
- Connectivity constraints

PROPOSED SYSTEM

5. Final Route Construction

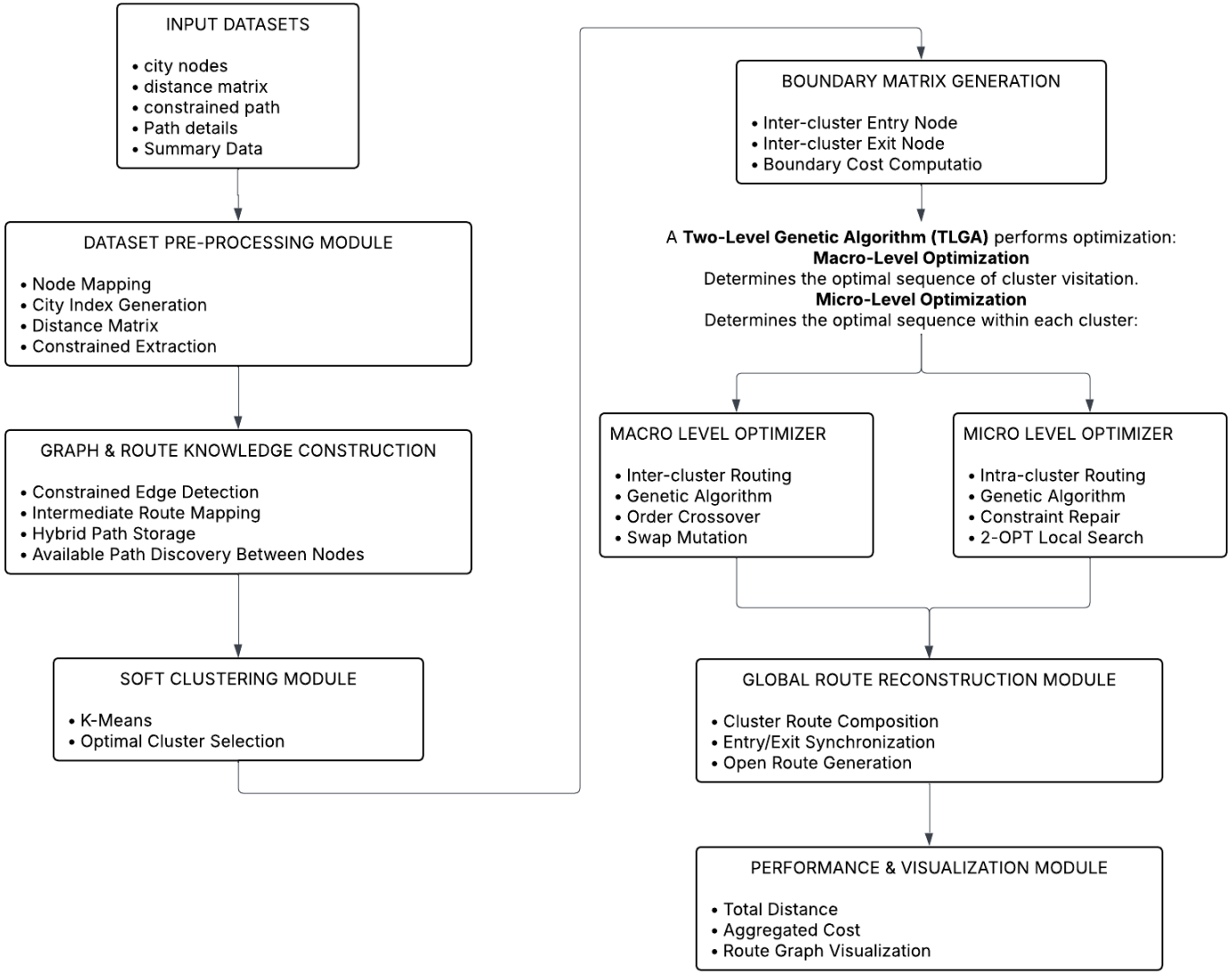
Optimized intra-cluster and inter-cluster routes are merged to generate the final delivery path.

Key Innovation

Unlike conventional TSP systems that assume direct point-to-point movement, the proposed system:

- Identifies actual feasible paths between locations
- Evaluates multiple route alternatives
- Optimizes based on real transportation connectivity
- Supports open-route agricultural delivery

SYSTEM ARCHITECTURE



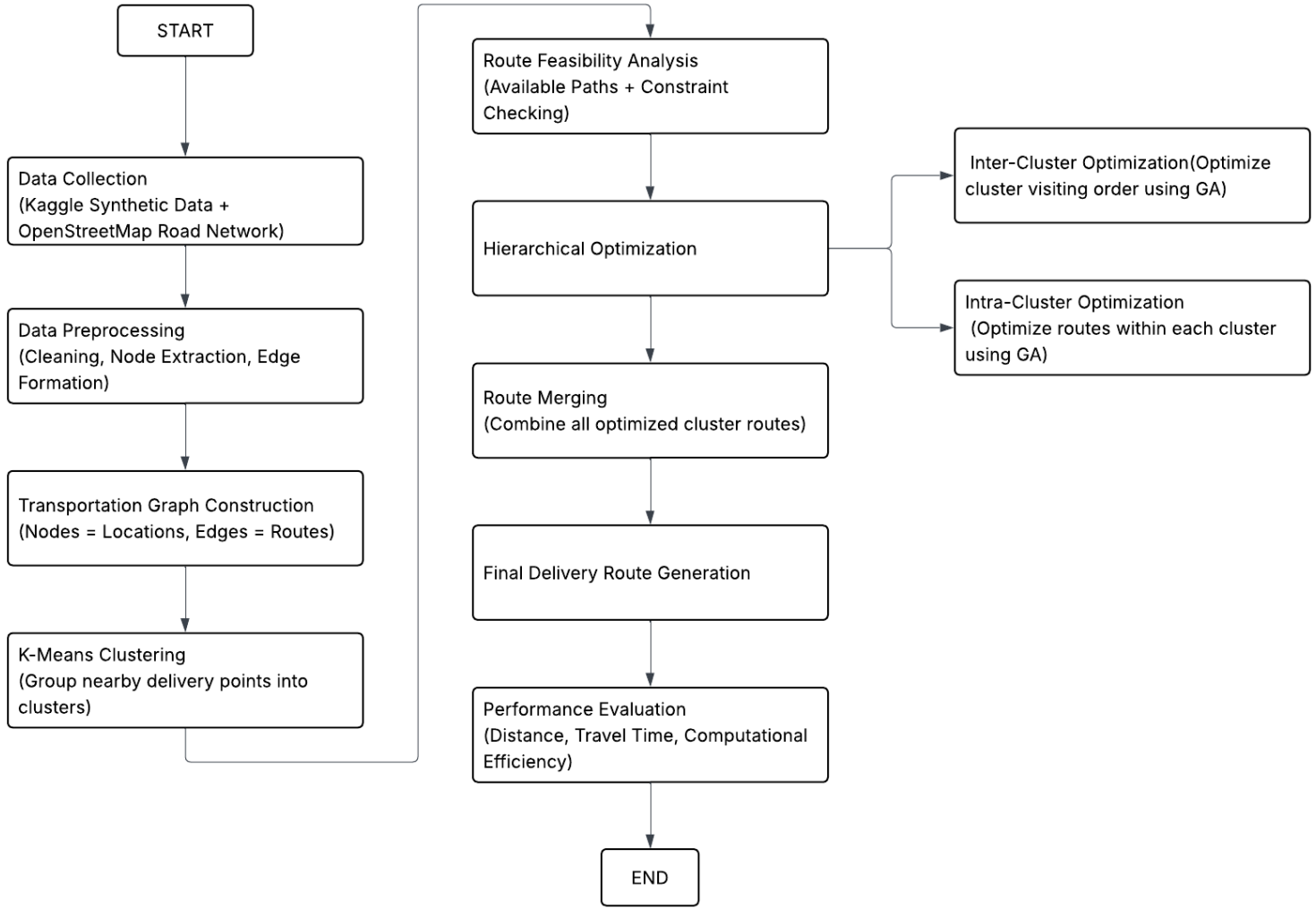
SYSTEM ARCHITECTURE

Input Layer: The system begins by loading city nodes, unconstrained distances, road constraints, and predefined path information.
Route Intelligence Layer: Unlike traditional TSP systems, the model first checks **available road paths between cities**, including intermediate cities if direct travel is constrained.
Clustering Layer: Using **K-Means clustering**, geographically close nodes are grouped to reduce search complexity.

Optimization Layer
Macro Level: Optimizes **cluster visitation order**.
Micro Level: Optimizes **node-level routing inside each cluster**, while respecting:
•Must-pass intermediate nodes
•Distance and time objectives
•Open route constraints

Final Layer
All optimized routes are merged to generate the final agricultural delivery path.

METHODOLOGY



METHODOLOGY

The proposed system follows a hierarchical workflow for route optimization:

1. Collect transportation data from Kaggle(synthetic) and OpenStreetMap (Real world)
2. Preprocess data and build the transportation graph
3. Apply K-Means clustering to group nearby nodes
4. Identify available paths and constrained routes between nodes
5. Perform inter-cluster optimization using Genetic Algorithm
6. Perform intra-cluster optimization using Genetic Algorithm
7. Merge all optimized routes into the final delivery path
8. Evaluate route performance using distance and time metrics

DATASET DESCRIPTION

• **Dataset Source:** The system uses synthetic datasets for large-scale testing and real-world road network data obtained from OpenStreetMap to reflect realistic routing conditions.

• **Number of Records:** The synthetic dataset consists of approximately **1,000 nodes**, while the real-world dataset contains around **100 nodes** representing actual locations.

• **Attributes / Features Used:** Each data point includes spatial coordinates (latitude and longitude), which are used to compute distances and construct routing paths.

• **Data Preparation Steps:** The dataset is preprocessed by cleaning missing values, normalizing coordinates, and converting location data into graph representations suitable for clustering and optimization.

TOOLS & TECHNOLOGIES

- **Programming Language:** The system is implemented using Python for its strong support in scientific computing and optimization tasks.
- **Frameworks:** Libraries such as NumPy, Pandas, pygad, Matplotlib
- **Database:** The system operates on structured datasets stored in file-based formats, eliminating the need for a dedicated database system.
- **Development Environment:** Implementation and testing are carried out using Jupyter Notebook and Google Colab for efficient experimentation and visualization.
- **Dataset Source:** Synthetic datasets are generated for scalability testing, while real-world road network data is obtained from OpenStreetMap to ensure realistic evaluation.

MODULES

1. **Input & Preprocessing Module:** Handles dataset loading, cleaning, and conversion of node coordinates into a structured format suitable for routing and graph representation.

2. **Clustering Module:** Applies K-Means Clustering to group geographically close nodes, reducing the complexity of the routing problem.

3. **Optimization Module:** Implements a Two-Level Genetic Algorithm to optimize inter-cluster and intra-cluster routes efficiently.

4. **Route Construction Module:** Integrates optimized cluster sequences and local routes to generate the final complete delivery path.

5. **Visualization & Analysis Module:** Displays routing results, cluster formations, and performance metrics for better interpretation and evaluation.

ALGORITHM , MODEL USED & IMPLEMENTATION SCREENS

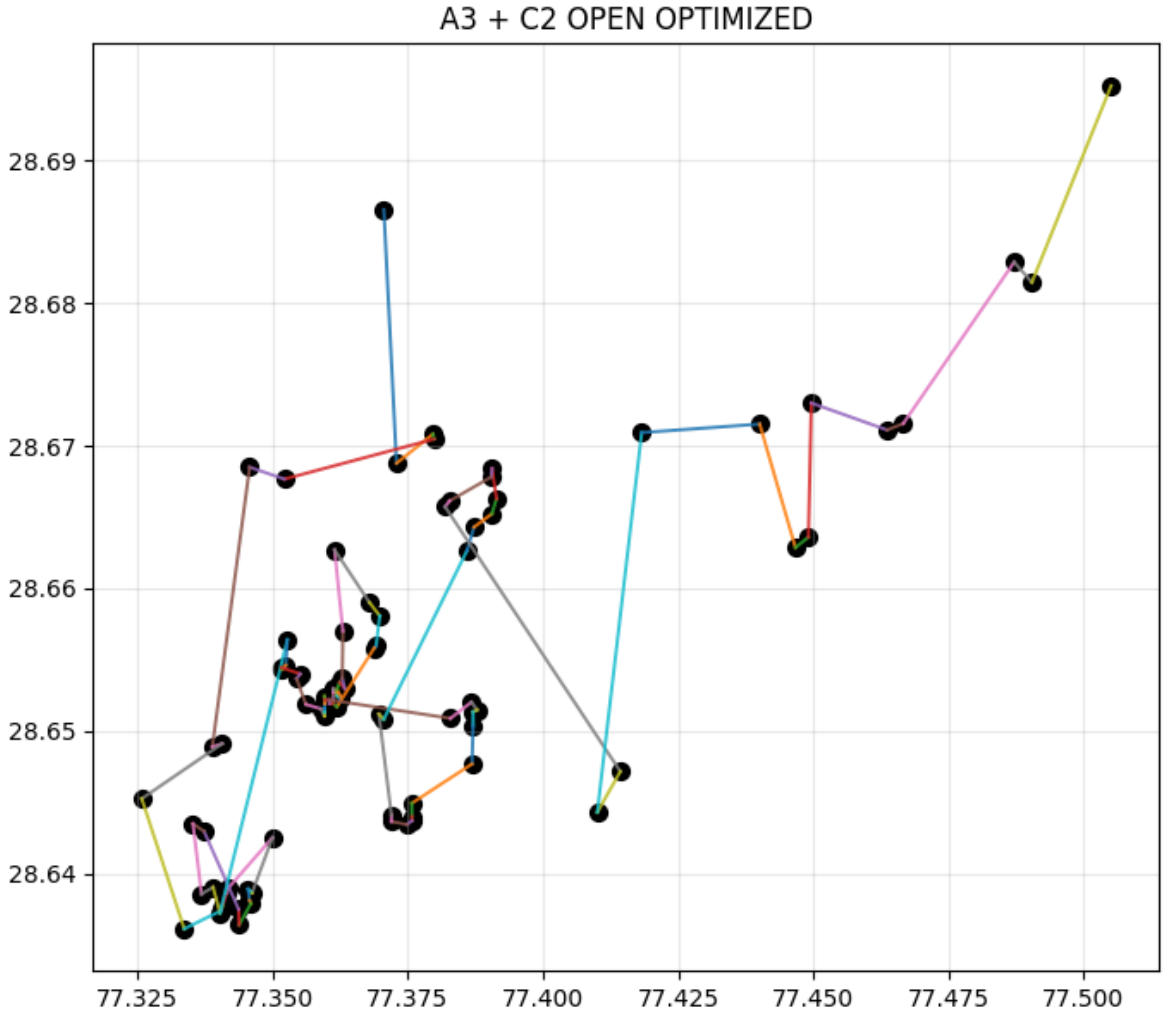
Algorithms Used:

1. K-Means Clustering
Used for grouping geographically close delivery points.

2. Two-Level Genetic Algorithm
Used for route optimization at two levels:
•**Macro Level:** Cluster sequencing
•**Micro Level:** Node-level routing

3. 2-OPT Optimization
Used for local route improvement.

ALGORITHM , MODEL USED & IMPLEMENTATION SCREENS



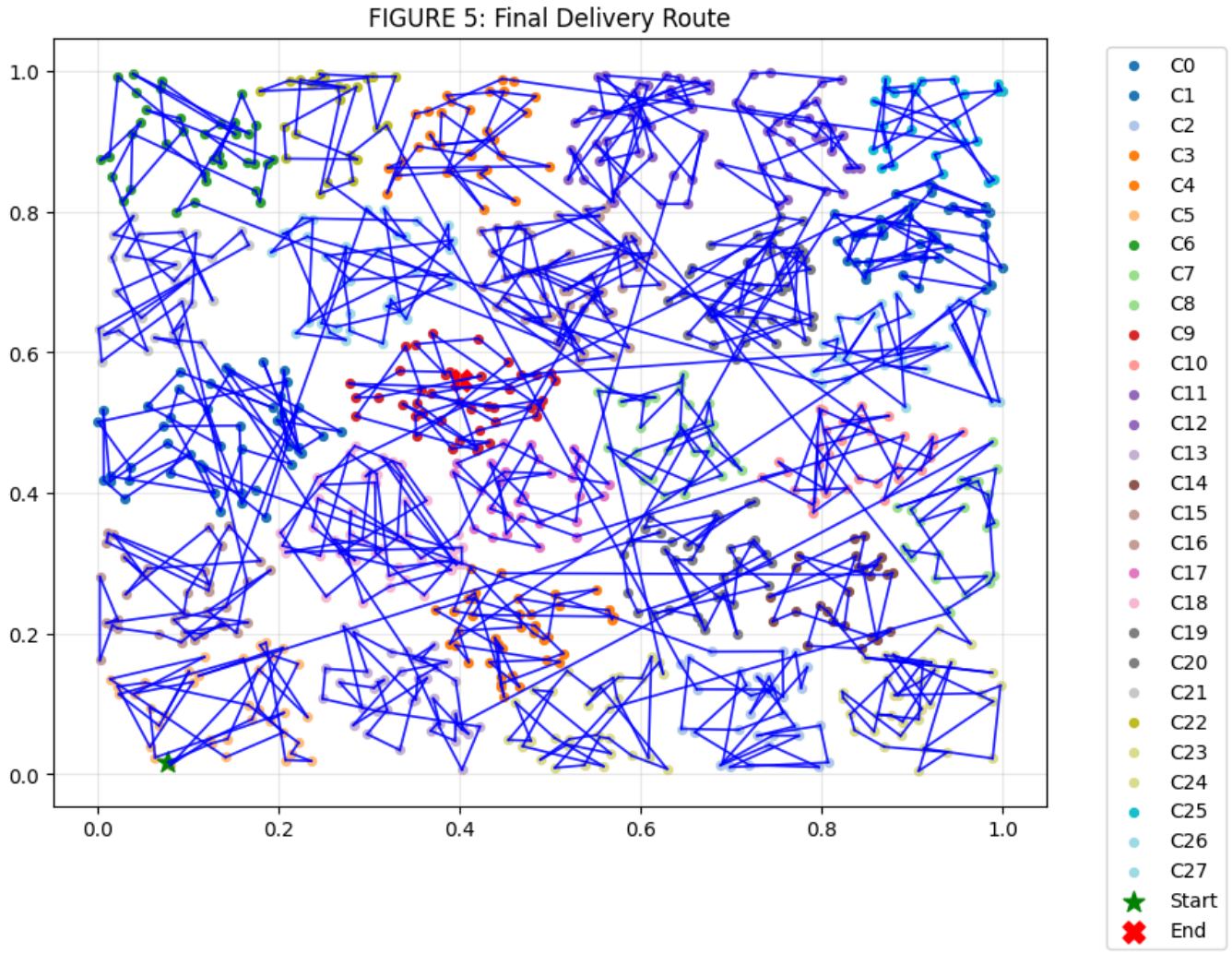

```
=== GLOBAL METRICS (OPEN) ===
Distance (D): 70.760
Time (T): 70.760
Aggregated cost: 70.760
Clusters visited: 10
End cluster: 4
Route length: 100

Cluster order: [9, 7, 2, 0, 3, 6, 5, 8, 1, 4]

Route [0, 56, 59, 60, 55, 54, 36, 35, 99, 49, 44, 41, 42, 39, 45, 40, 48,
```

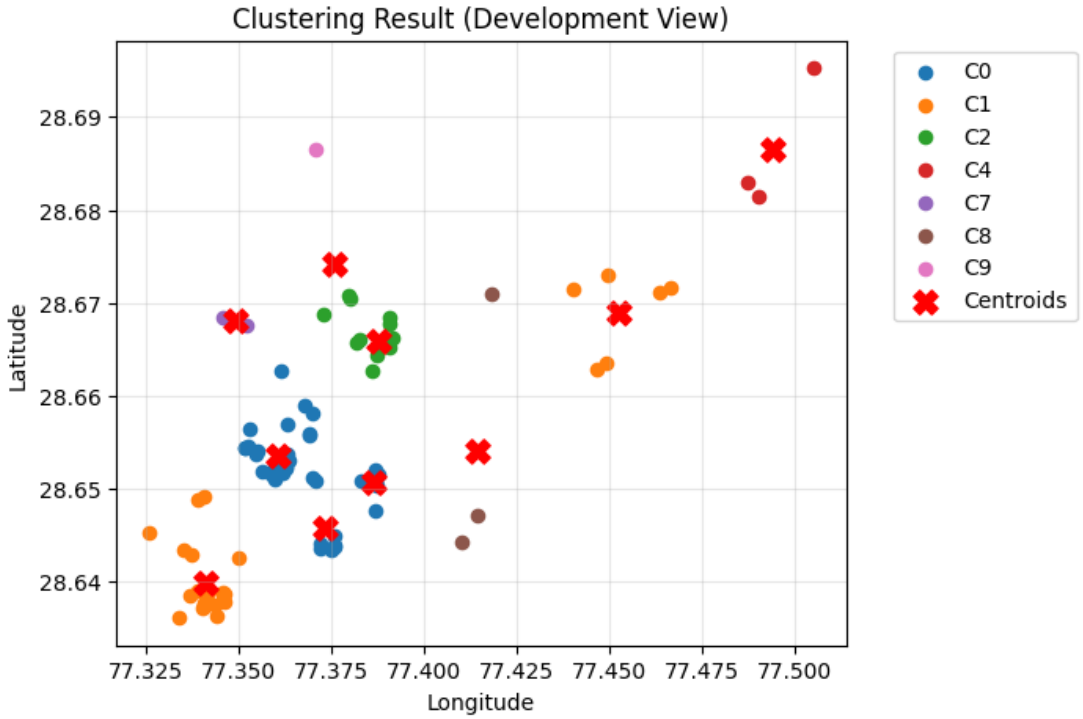
ALGORITHM , MODEL USED & IMPLEMENTATION SCREENS

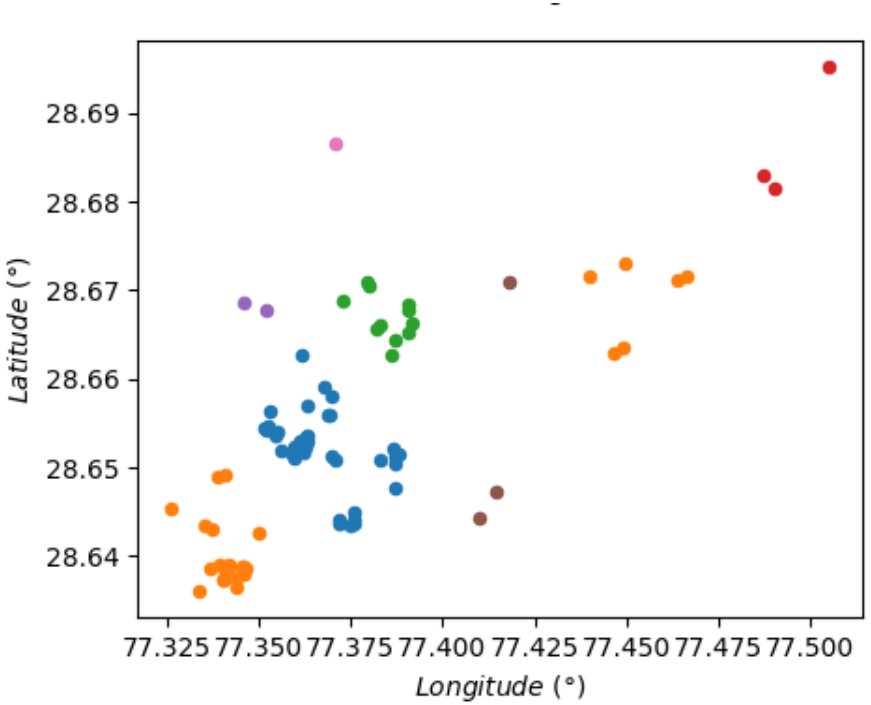
```
{'total_distance': np.float64(62.28426917763631),
 'intra_distance': np.float64(61.92124023487806),
 'inter_distance': np.float64(0.3630289427582463),
```



DEMO FLOW/ WORKING MODEL

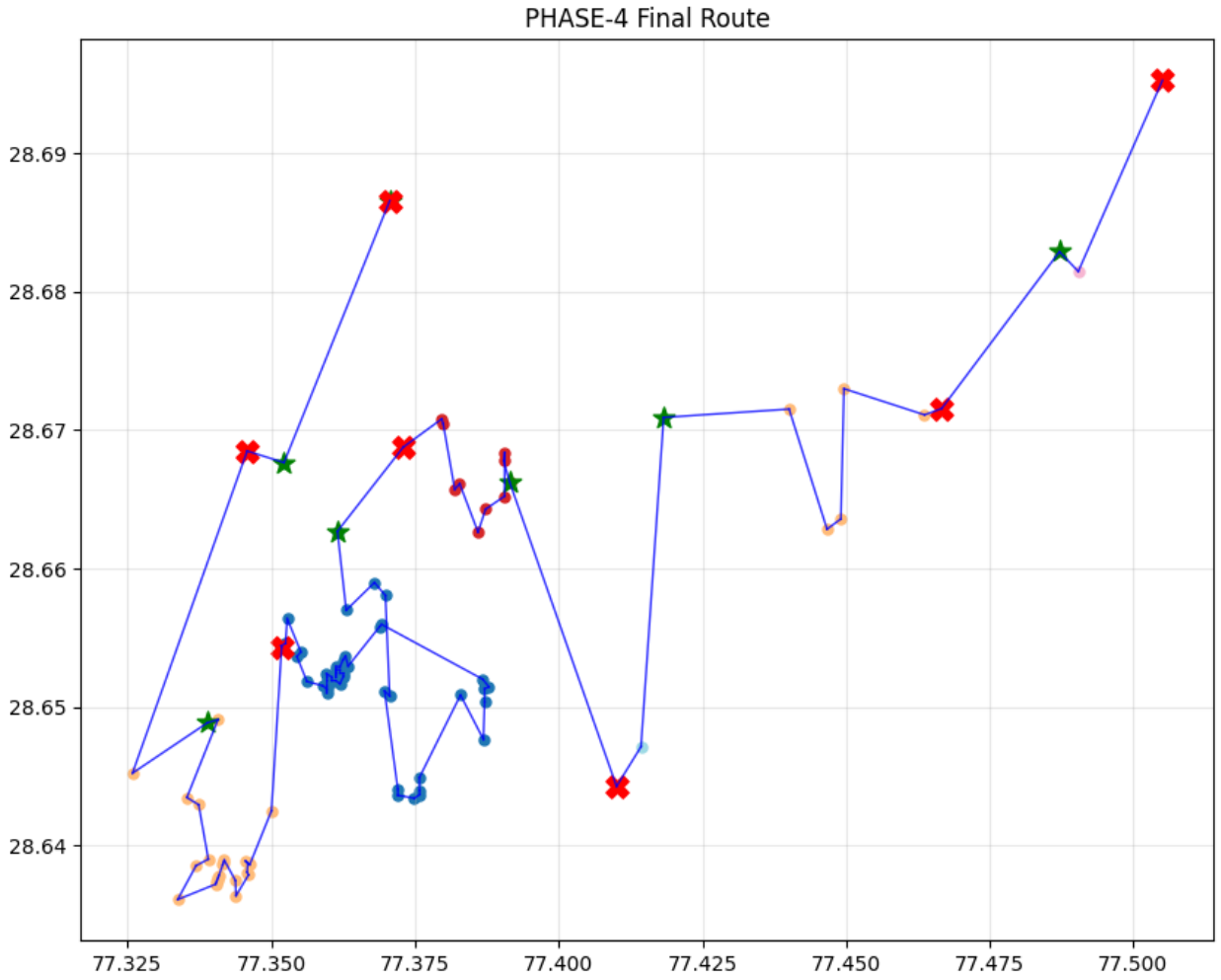
```
PHASE-1 DATASET READY ✓
Nodes: 100
Constrained edges: 6608
Hybrid edges: 9900
Points loaded for clustering: (100, 2)
Selected K (medium range 8–10) = 10
KMeans clustering complete with medium K.
Clusters: 10
DBSCAN refinement: 15 noise points
Reassigning 15 noise to nearest KMeans centroid...
Final clusters = 7
```






```
Cluster nodes built for 7 clusters:
  Cluster 0: 50 nodes
  Cluster 1: 30 nodes
  Cluster 2: 11 nodes
  Cluster 4: 3 nodes
  Cluster 7: 2 nodes
  Cluster 8: 3 nodes
  Cluster 9: 1 nodes
Start node = 0, belongs to cluster C9
```

DEMO FLOW/ WORKING MODEL




```
=== FINAL METRICS ===
distance: 138.69612998000002
time: 138.69612998000002
cost: 138.69612998000002
clusters_visited: 7
start_cluster: 9
end_cluster: 4
route_length: 100
execution_time_sec: 14.560195922851562

=== CLUSTER ORDER ===
[np.int64(9), 8, 2, 0, 7, 1, 4]

=== ENTRY NODES ===
{np.int64(9): 0, 8: 4, 2: 63, 0: 67, 7: 55, 1: 36, 4: 6}

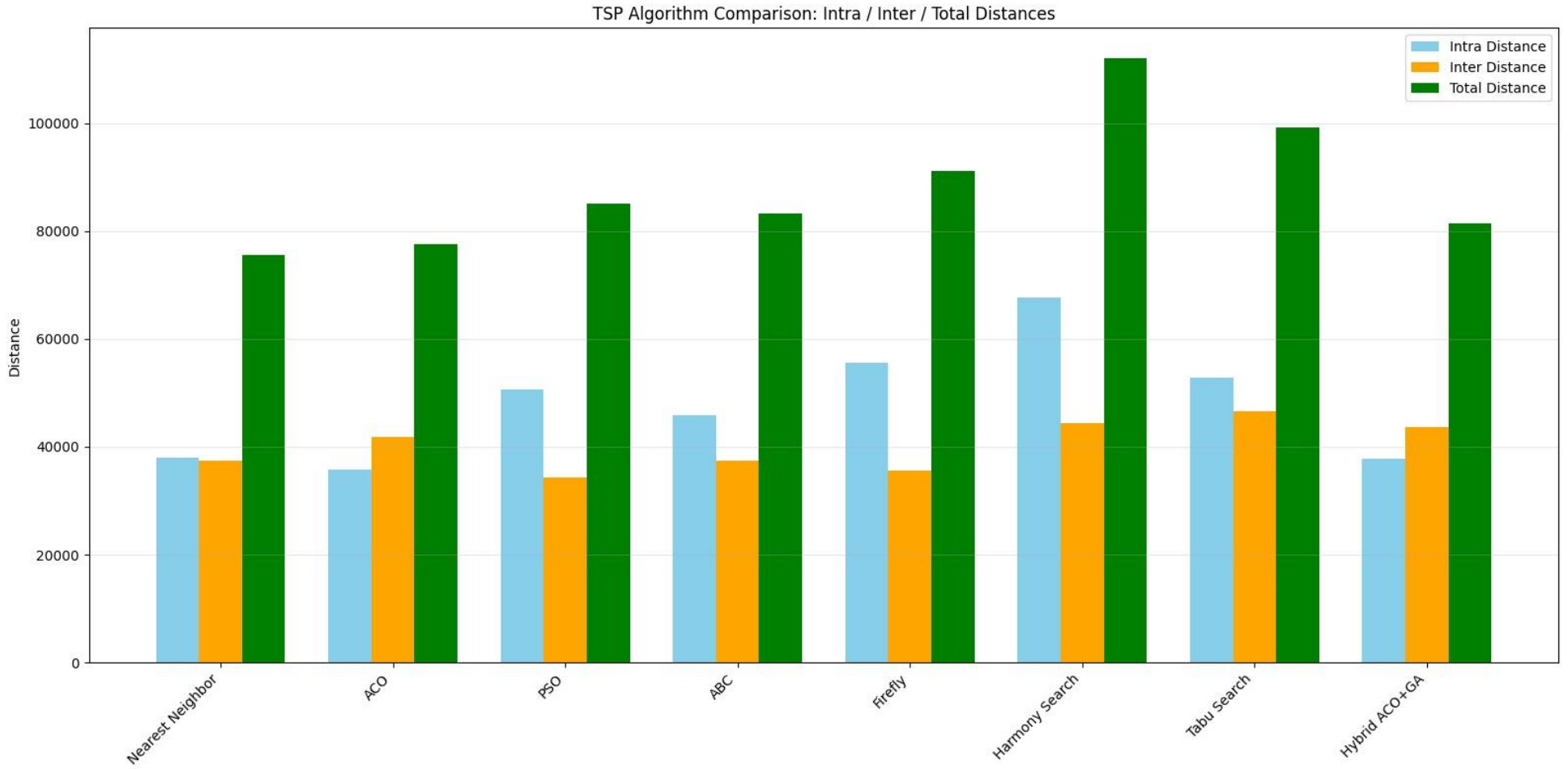
=== EXIT NODES ===
{np.int64(9): 0, 8: 11, 2: 56, 0: 52, 7: 54, 1: 8, 4: 1}

=== GLOBAL ROUTE LENGTH ===
100

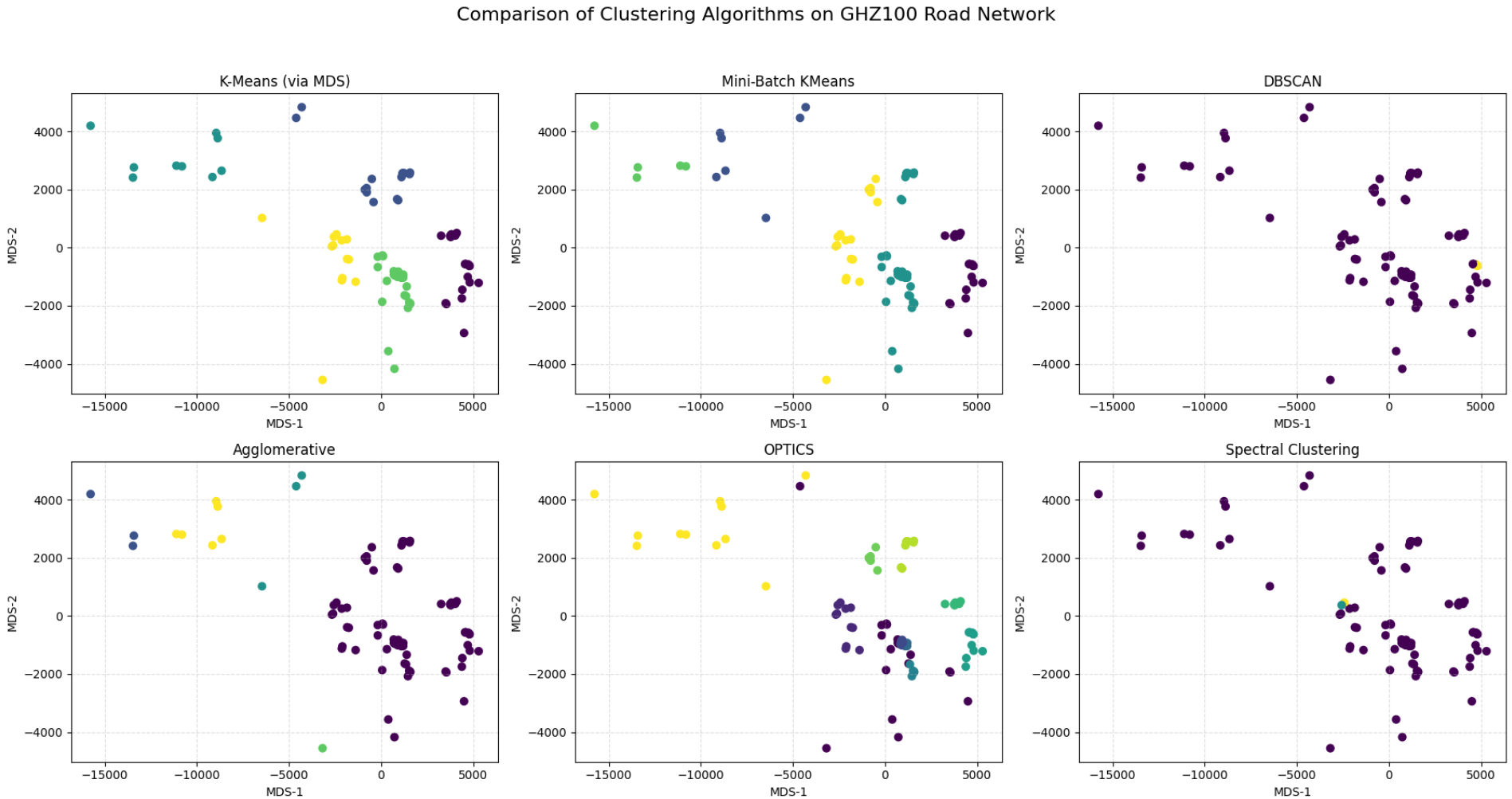
=== ROUTE NODES ===
[0, 55, 54, 99, 36, 35, 43, 47, 37, 38, 49, 46, 44, 41, 42, 39, 45, 40, 48, 29, 33, 28, 32, 31,

PHASE-4 COMPLETED ✓
```

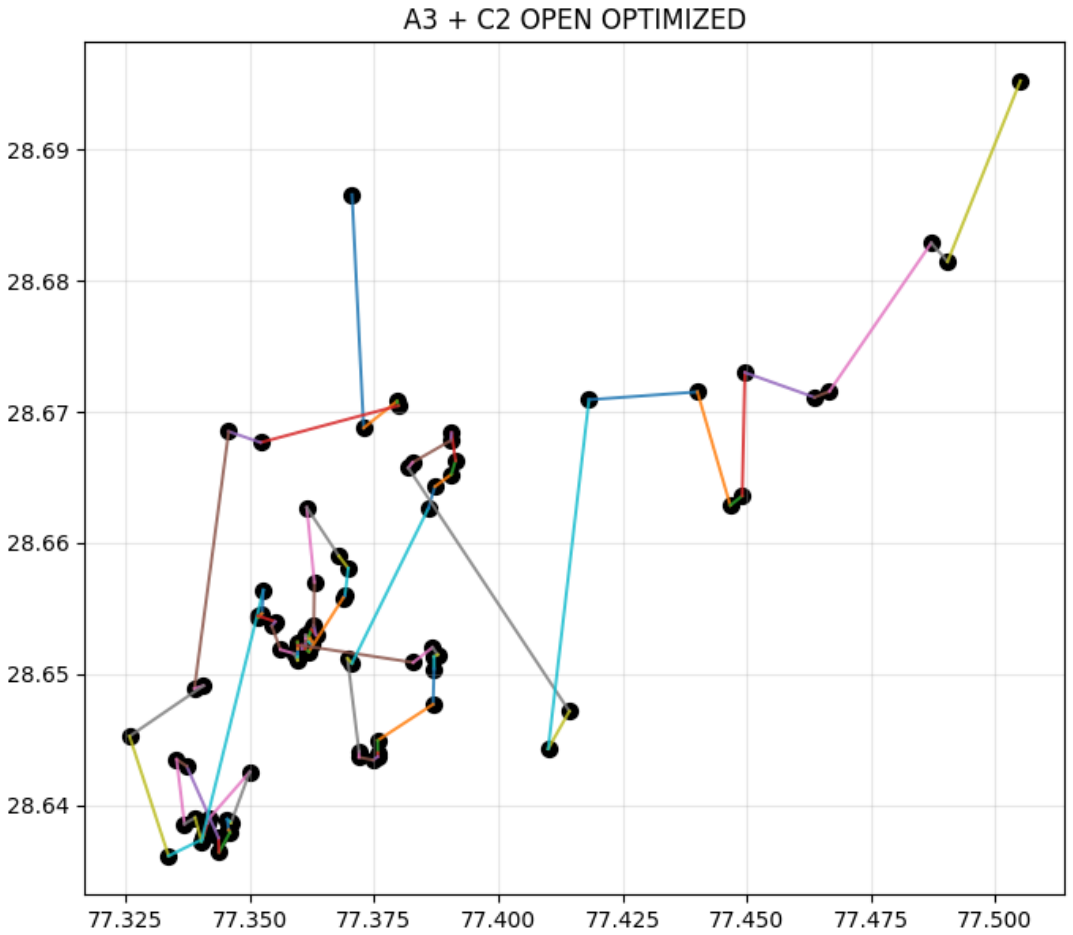
EXPERIMENTAL RESULTS

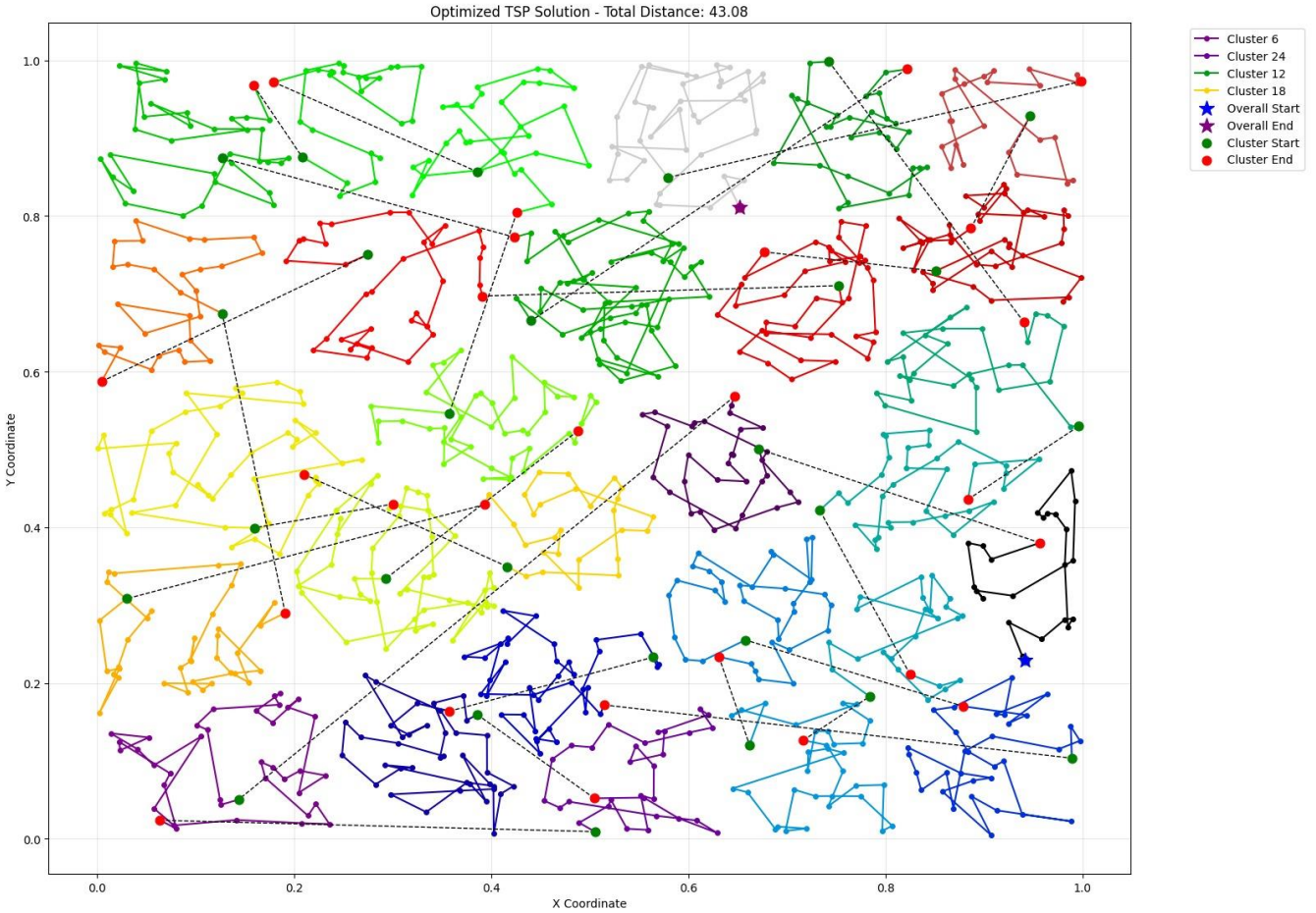


EXPERIMENTAL RESULTS



EXPERIMENTAL RESULTS



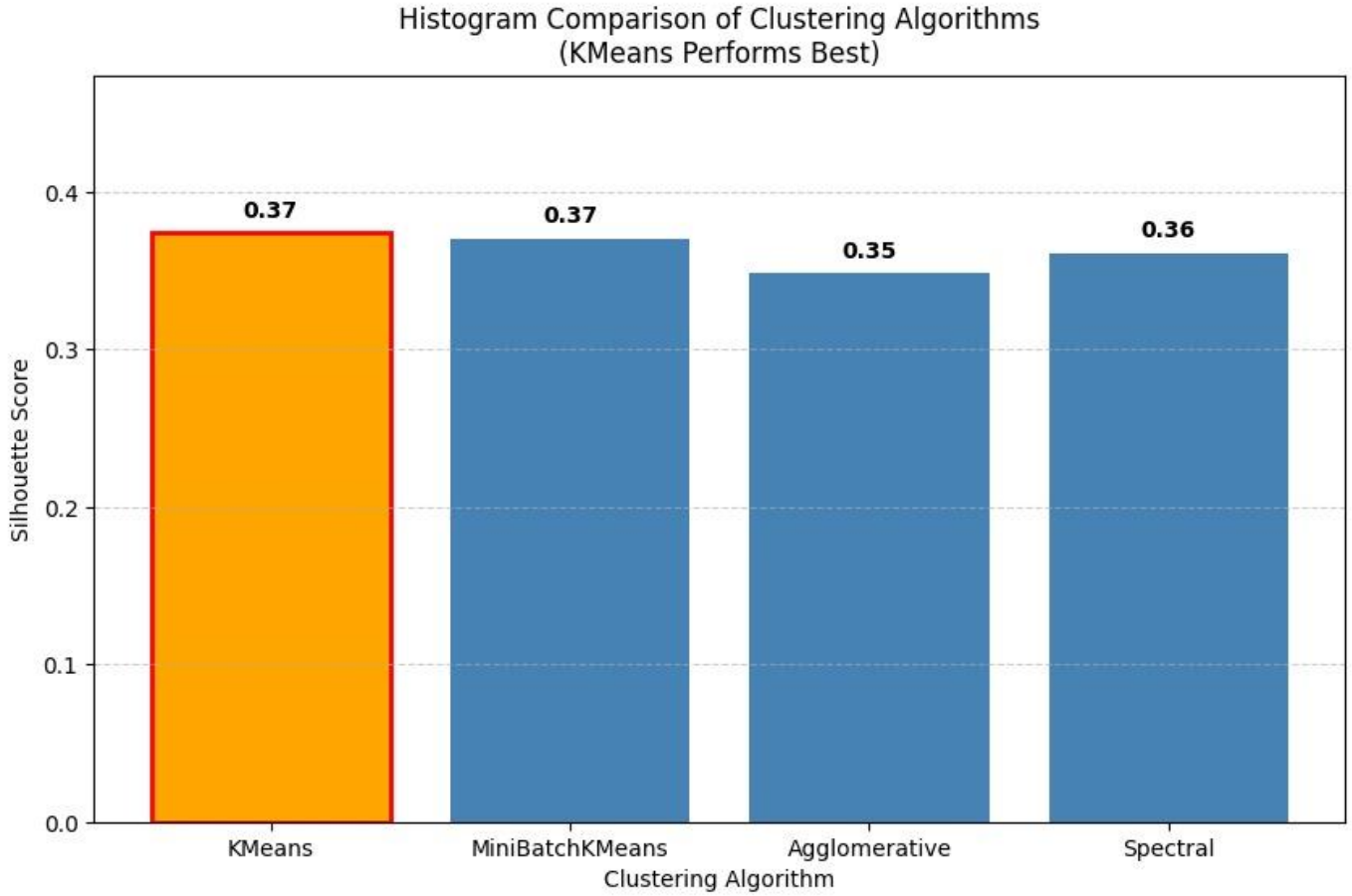


Inter-cluster Path Sequence

Segment 1:	Cluster 9 → Cluster 8 → Cluster 6 → Cluster 24 → Cluster 14 → Cluster 5
Segment 2:	Cluster 25 → Cluster 21 → Cluster 3 → Cluster 15 → Cluster 11 → Cluster 28
Segment 3:	Cluster 12 → Cluster 16 → Cluster 7 → Cluster 23 → Cluster 4 → Cluster 10
Segment 4:	Cluster 19 → Cluster 2 → Cluster 18 → Cluster 17 → Cluster 22 → Cluster 27
Segment 5:	Cluster 20 → Cluster 1 → Cluster 26 → Cluster 13

EXPERIMENTAL RESULTS

No.	Clustering Techniques	Silhouette Score
1.	K-Means	0.378
2.	MiniBatch K-Means	0.373
3.	Agglomerative	0.358
4.	Spectral	0.357
5.	DBSCAN	NA
6.	OPTICS	NA



PERFORMANCE METRICS

1. Total Travel Distance (TTD)

Measures the **overall distance traveled** while visiting all delivery locations.

- Primary evaluation metric of the proposed routing system.
- Lower TTD indicates better route optimization and reduced transportation cost.
- Used to compare the efficiency of different routing strategies.

2. Computational Efficiency

Measures the **processing time required** by the algorithm to generate optimized routes.

- Evaluates how quickly the system produces solutions.
- Important for real-time logistics and delivery planning.
- Lower execution time indicates better computational performance.

3. Scalability

Measures the system's ability to maintain performance as the number of delivery nodes increases.

- Evaluates performance on larger datasets.
- Tests robustness under real-world transportation workloads.
- Indicates whether the optimization framework can handle expanding delivery networks.

PERFORMANCE METRICS

No.	Clustering Techniques	Silhouette Score
1.	K-Means	0.378
2.	MiniBatch K-Means	0.373
3.	Agglomerative	0.358
4.	Spectral	0.357
5.	DBSCAN	NA
6.	OPTICS	NA

Silhouette Score of Clustering Algorithms

Algorithms	**TTD**
TLGA (1000 points) unconstrained	43.06
TLGA (1000 points) constrained	62.28
Proposed TLGA (real world)	70.760

Algorithms and Total Travelled Distance

RESULTS & DISCUSSION

Key Observations

The proposed framework:

- Successfully handled constrained road graphs
- Found feasible intermediate routes
- Reduced unnecessary cluster transitions
- Generated stable open-route solutions

Comparison with Existing Methods

Method	Scalability	Constraint Handling
Classical GA	Medium	No
ACO	Medium	Limited
PSO	Medium	No
Proposed TLGA	High	Yes

INNOVATION & CONTRIBUTION

Key Innovations : Path-aware route optimization, Soft clustering with boundary routing, Constraint repair mechanism, Open-route agricultural delivery model

Individual Contribution Role mentioned here

Ninaad Saxena – Testing and Implementation of Genetic Algorithm on Synthetic dataset, performing final route optimization for TLGA and Research paper drafting.

Ritvik Srivastava – Implementation and Visualization of TLGA on synthetic and real-world dataset along with final comparison of Algorithms.

Utkarshi Birla – Testing, analysis, Comparison (silhouette score) and visualization of different Clustering algorithms and Research paper drafting.

Mohak Kansal – Testing and implementation of Various Metaheuristic Algorithms along with comparison and visualization.

RESEARCH PAPER DETAILS

- **Paper Title** : Leveraging the Genetic Algorithm for Optimizing TSP in Agricultural Seed and Product Transportation

- **Authors :** Ninaad Saxena, Ritvik Srivastava, Utkarshi Birla, Mohak Kansal, Dr. Kanika Garg (Guide)

- **Conference/Journal Name:** 2nd International Conference on Intelligent Computing and Sustainable Innovation in Technology (IC-SIT) 2026

- **Date of Conference (if Paper Published in the conference) : N.A.**

- **Indexing (Scopus/WoS) : IEEE Xplore , Scopus**

RESEARCH PAPER DETAILS

First page of the submitted paper

Leveraging the Genetic Algorithm for Optimizing TSP in Agricultural Seed and Product Transportation

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***Abstract*—Agricultural logistics route optimization suffers significantly from computational complexities owing to the existence of dispersed delivery locations, irregular and sparse road networks and open-route delivery conditions. Inefficient routing increases travel time, fuel consumption and operational costs. To address these challenges, this paper proposes a hierarchical optimization framework that models the routing of agriculture logistics through a Clustered Traveling Salesman Problem (CTSP), using both the K-Means clustering algorithm and a Two-Level Genetic Algorithm (TLGA), to independently optimize the sequencing of inter-cluster traveling and intra-cluster traveling within open-route constraints. The proposed approach is tested on a large-scale synthetic dataset with a 1000-node problem and a real-world 100-node road network dataset of Ghaziabad obtained from OpenStreetMap. The results indicate that the TLGA framework consistently achieves lower total travel distance (TTD) compared to both single-level optimization and other metaheuristic algorithms. The reduced routing cost directly translates to improved delivery efficiency and reduced fuel consumption, making the proposed method suitable for real-world agricultural logistics applications.**

***Keywords*—Agricultural logistics, Traveling Salesman Problem, clustered TSP, two-level genetic algorithm, K-Means clustering, open-route optimization, road network–constrained routing.**

I. Introduction

Efficient logistics are important for a sustainable and profitable agriculture economy. In rural supply chains without enough technological resources and sufficient transport infrastructure, seeds, products, fertilizers and harvests must be transported over enormous distances among dispersed farms. Thus inefficient route planning increases the distance traveled, fuel consumption and total operational costs which in turn leads to increased port-harvest losses. Such ineffectiveness has implications in terms of profitability and access to market prospects for farmers.

Many agricultural transportation challenges can be considered as versions of the Travelling Salesman Problem (TSP), an optimization problem that finds an ideal visiting sequence to minimize the total journey cost. The TSP itself is a well-known NP-hard problem, hence it is computationally too expensive for finding an optimal solution for large-scale real-world logistics problems [1]. Therefore, heuristic and meta-heuristic methods have been extensively applied to find near optimal solutions within acceptable computing limits. Among them, Genetic Algorithms (GAs) have shown a remarkably good performance due to their robustness, adaptability and effectiveness in searching huge and complex search spaces [2], [6].

Traditional formulations of the Traveling Salesman Problem (TSP) do not take into account several real constraints that occur in agricultural logistics. Most of the prior literature assumes either fully connected graphs or Euclidean distance models where the trip expenses are modeled as straight line lengths between sites. However, such assumptions do not realistically represent rural transportation networks. In reality rural road networks often have limited connectivity, constrained in degree and path-dependent in travel, where travel between two sites generally goes through planned intermediate junctions, not direct links [15],[21].

Additionally, agricultural distribution routes are often open-ended, with vehicles ending their trips at markets, warehouses or collection points rather than returning to the starting station. This operating characteristic is quite different from the traditional closed-loop TSP formulation. Hence, in the presence of such domain-specific constraints, straightforward routing algorithms such as the Nearest Neighbour Algorithm may not produce optimal or feasible solutions.

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PUBLICATION PROOF

- Submission Proof:

Fwd: 2nd International Conference on Intelligent Computing and Sustainable Innovation in Technology (IC-SIT) 2026 : Submission (359) has been created. Inbox ×

NINAAD SAXENA (RA2211026030057)
to me

Apr 25, 2026, 2:01 PM (11 days ago)

---------- Forwarded message ---------
From: **Microsoft CMT** <noreply@msr-cmt.org>
Date: Sat, Apr 4, 2026 at 11:06 AM
Subject: 2nd International Conference on Intelligent Computing and Sustainable Innovation in Technology (IC-SIT) 2026 : Submission (359) has been created.
To: <ns5778@srmist.edu.in>

Hello,

The following submission has been created.

Track Name: ICSITSU2026

Paper ID: 359

Paper Title: Leveraging the Genetic Algorithm for Optimizing TSP in Agricultural Seed and Product Transportation

Abstract:
Agricultural logistics routing involves significant computational challenges due to geographically dispersed delivery locations, sparse road connectivity, and open-route delivery requirements. Inefficient routing in such environments leads to increased travel time, fuel consumption, and operational costs. To address these challenges, this paper proposes a hierarchical optimization framework that formulates agricultural transportation routing as a Clustered Traveling Salesman Problem (CTSP). The framework integrates K-Means clustering for spatial decomposition with a Two-Level Genetic Algorithm (TLGA) that separately optimizes inter-cluster sequencing and intra-cluster routing under open-route constraints. The proposed approach is evaluated on a large-scale 1,000-node synthetic dataset and a 100-node real-world road network from the Ghaziabad region derived from OpenStreetMap, capturing realistic road topology and connectivity constraints. Experimental results demonstrate that the TLGA-based framework consistently achieves lower total travel distance compared to single-level and alternative metaheuristic approaches. The reduced routing cost directly translates to improved delivery efficiency and reduced fuel consumption, highlighting the suitability of the proposed method for real-world agricultural logistics planning.

Created on: Sat, 04 Apr 2026 05:35:57 GMT

Last Modified: Sat, 04 Apr 2026 05:35:57 GMT

LIMITATIONS & CHALLENGES

Limitations

1. **Data Constraints:** The paper uses only two datasets—a synthetic 1,000-node set and a real-world 100-node network from Ghaziabad—lacking standardized benchmarks and diversity in geography, road conditions, and crop patterns. The real-world data is limited to one district, which may not represent broader rural agricultural contexts.
2. **Scalability:** While the TLGA performs well on 1,000 nodes, real-world national or regional agricultural supply chains can involve thousands of delivery points. The computational cost of K-Means + TLGA at very large scales isn't validated, leaving scalability to massive problems an open question.
3. **Accuracy Limitations:** TLGA provides near-optimal routes but doesn’t guarantee the best solution and only minimizes travel distance, ignoring real-world factors like vehicle capacity, time, road quality, fuel costs, and load weight.

LIMITATIONS & CHALLENGES

Challenges Faced

1. **Dataset Limitations:** No standardized benchmarks exist for agricultural logistics with realistic road constraints, so we built our own dataset from OpenStreetMap, which may have incomplete or inaccurate road information.

2. **Technical Issues:** Modeling the road network as a ***degree-constrained graph*** significantly complicates routing compared to standard Euclidean TSP. Algorithms like DBSCAN and OPTICS couldn't produce valid silhouette scores, indicating they failed to form meaningful clusters on this data

3. **Implementation Challenges:** Implementing a two-level genetic algorithm in a web system poses real-time performance challenges, requiring a balance between solution quality and quick responses. Ensuring fair comparison across algorithms with open-route constraints also demanded careful framework design..

APPLICATIONS

System Deployment / Real-Time Usage

Web-Based Decision-Support System: The system enables users to enter delivery locations and receive optimized routes instantly via a user-friendly web interface, making complex optimization accessible without requiring algorithmic knowledge.

API : An API layer would allow third-party agri-logistics platforms, ERP systems, or government supply chain portals to plug into the optimization engine, making the framework reusable at an ecosystem level rather than just a standalone tool.

APPLICATIONS

Industry Use Case

The core industry use case is **agricultural seed and product transportation** in rural and semi-urban India — specifically modeled on the Ghaziabad region. Concretely, this covers:

Input distribution — delivering seeds, fertilizers, and pesticides from warehouses to dispersed farms.

Output collection — transporting harvested produce from farms to markets or aggregation centers (which is why open-route formulation matters — vehicles end at the market, not back at the depot).

Rural supply chain efficiency — reducing fuel costs, post-harvest losses from delays, and operational overhead for farmer cooperatives or government agencies managing last-mile agricultural logistics.

DEPLOYMENT

- Architecture (Cloud / Local): Local
- Tech stack – Already mentioned
- We plan to deploy our proposed framework as an open-source python package/ Library after the publishing of our research paper.

CONCLUSION & FUTURE WORK

Key Achievements

- The paper successfully proposes and validates a **hierarchical optimization framework** that combines **K-Means clustering** with a **Two-Level Genetic Algorithm (TLGA)** to solve the Clustered Traveling Salesman Problem (CTSP) for agricultural logistics. The key achievements are:
- The TLGA **outperformed all competing metaheuristics** — including ACO, PSO, BCO, Firefly Algorithm, Tabu Search, Harmony Search, and hybrid ACO-GA — in terms of total travel distance on both datasets.
- On the **1,000-node synthetic dataset**, the framework achieved a minimum total travel distance of **43.08**, with intra-cluster distance of **35.24** and inter-cluster distance of only **7.83** — showing that K-Means clustering effectively grouped geographically close nodes, keeping macro-level navigation overhead very low.
- On the **100-node real-world Ghaziabad dataset** with actual road network constraints (sparse connectivity, degree-constrained graphs), the TLGA maintained a total travel distance of **70.760**, proving robustness under realistic conditions.
- The framework was successfully deployed as a **web-based decision-support system**, demonstrating practical usability beyond just academic benchmarking.

CONCLUSION & FUTURE WORK

Final Outcome

The proposed framework directly addresses the core inefficiencies in rural agricultural logistics — excessive travel distance, high fuel consumption, and delayed deliveries. By reducing routing cost, the system improves delivery efficiency, lowers post-harvest losses, and enhances market accessibility for farmers. The open-route formulation makes it especially realistic, since vehicles terminate at markets or collection centers rather than returning to a depot — matching how agricultural distribution actually works on the ground.

CONCLUSION & FUTURE WORK

Future Work

1. **Enhancements:** We plan to integrate **dynamic data sources** such as real-time traffic conditions, weather information, and seasonal road accessibility into the optimization pipeline. This would shift the system from static route planning to **adaptive, real-time route optimization** — far more useful in practice where road conditions change frequently in rural areas.

2. **Scalability:** The framework will be extended to **multi-vehicle and cooperative transportation models**, better representing shared logistics systems used by farmer groups (FPOs) and rural aggregators where multiple vehicles operate simultaneously and routes need to be coordinated. Further validation is also planned across **larger and more geographically diverse regions** beyond Ghaziabad to test robustness under varying infrastructure and environmental conditions.

3. **Deployment Ideas:** The existing web-based system is a proof of concept. Future deployment ideas include scaling it into a full **mobile-accessible platform** for field agents, integrating it with **government agricultural portals** or FPO management systems via an API, and potentially coupling it with **IoT-based vehicle tracking** for real-time route adjustment during delivery.

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THANK YOU